

MEMO.

WTW158 Calculus

Class Test 7 (Mo)

29 - 2 June 2023

Time: 15 min

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Initials:

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Student number:

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Use a pen to complete the test. The test is printed on both sides of this paper.

QUESTION 1 [1,1,1]

Find the most common anti-derivative f of the function given.

(1.1) $F(x) = \sin(x)$

$$f(x) = -\cos x + k$$

Test: $\frac{d}{dx} (-\cos x + k)$
 $= -\frac{d}{dx} (\cos x + k)$
 $= -(-\sin x + 0) = \sin x.$

(1.2) $F(x) = \frac{1}{x}$

$$f(x) = \ln x + k$$

(1.3) $F(x) = \frac{2}{1+x^2}$

$$f(x) = 2 \arctan x + k.$$

QUESTION 2 [5]

Consider $f(x) = \ln(4 - x^2)$, with $f'(x) = \frac{-2x}{4 - x^2}$ and $f''(x) = \frac{-8 - 2x^2}{(4 - x^2)^2}$. The domain of f is $(-2, 2)$ and the graph of f has no horizontal asymptotes.

Sketch the graph of f by considering all intercepts, symmetry, vertical asymptotes, intervals of increase and decrease, local extreme values and concavity.

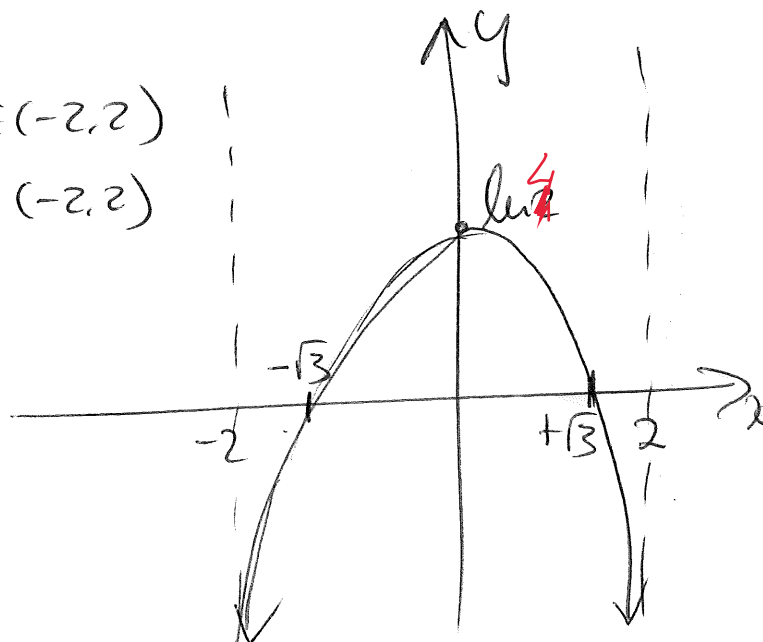
① Intercepts: $x=0 \Rightarrow y = \ln 4$
 $y=0 \Rightarrow 0 = \ln(4 - x^2)$
 $\Rightarrow e^0 = e^{\ln(4 - x^2)} = 4 - x^2$
 $\Rightarrow 4 - x^2 = 1$
 $\Rightarrow x^2 = 3 \Rightarrow x = \pm\sqrt{3}$.

② Symmetry:
 $f(-x) = \ln(4 - (-x)^2) = \ln(4 - x^2) = f(x) \Rightarrow f$ is even.

③ Vertical asymptotes:
 f is defined on $(-2, 2)$
 $\Rightarrow$ Test for vertical asymptotes at $x = -2$ and at $x = 2$
 $\left. \begin{array}{l} \lim_{x \rightarrow -2^+} \ln(4 - x^2) = -\infty \\ \lim_{x \rightarrow 2^-} \ln(4 - x^2) = -\infty \end{array} \right\} \Rightarrow x = -2 \text{ and } x = 2$
are vertical asymptotes.

④ $f'(x) = \frac{-2x}{4 - x^2} > 0 \Rightarrow x < 0$
 $\Rightarrow f$ increases on $(-2, 0)$ and decreases on $(0, 2)$

⑤ $f''(x) > 0$ for no x
 $f''(x) < 0$ for all $x \in (-2, 2)$
 $\Rightarrow f$ is concave down on $(-2, 2)$



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QUESTION 1 [1,1,1]

Find the most common anti-derivative f of the function given.

(1.1) $F(x) = \sec^2(2x)$

$$f(x) = \cancel{\ln(x) + k}$$

$$= \frac{1}{2} \ln(2x) + c$$

Test: $\frac{d}{dx}(\ln x + k)$

$$= \cancel{\sec^2 x + 0}$$

(1.2) $F(x) = a^x, a > 0$

$$f(x) = \frac{a^x}{\ln a} + c$$

Test: $\frac{d}{dx}(\frac{a^x}{\ln a} + c)$

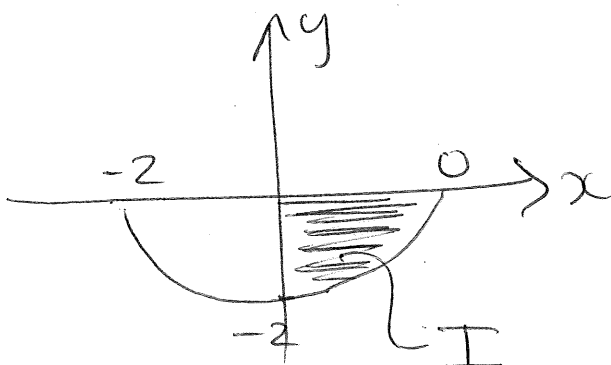
$$= \frac{a^x}{\ln a} \times \ln a = a^x.$$

(1.3) $F(x) = \frac{1}{1+x^2}$

$$f(x) = \arctan x + c$$

QUESTION 2 [2 marks]

Make use of a graph and areas to find the following integral: $I = \int_0^2 -\sqrt{4-x^2} dx$. Clearly indicate the region represented by I on the graph.



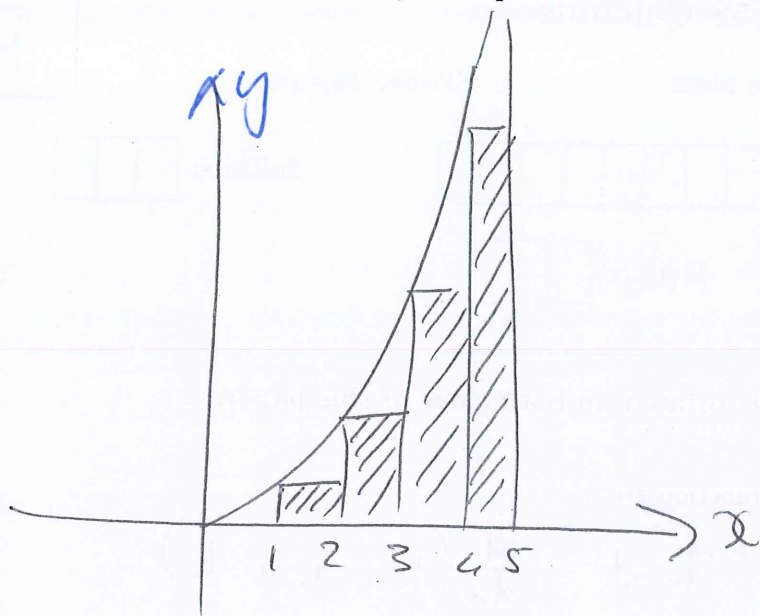
$$\text{Area} = \frac{1}{4} (\pi r^2)$$

$$= \frac{1}{4} (\pi \times 4) = \pi$$

$$\Rightarrow I = -\pi.$$

QUESTION 3 [3 marks]

Use a left hand sum with 4 sub-intervals to estimate the area between the graph of $y = x^2$, the x -axis and $x = 1$ and $x = 5$. Show all your steps and indicate the estimated area on a graph.



$$\Delta x = \frac{b-a}{n} = \frac{5-1}{4} = 1$$

$$\begin{aligned}\text{Area} &= 1 (f(1) + f(2) + f(3) + f(4)) \\ &= 1 + 4 + 9 + 16 \\ &= 30\end{aligned}$$

~~The area is an under-estimate of the actual area between the graph, the x -axis, $x=1$ and $x=5$.~~

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QUESTION 1 [1]

Write down the most common anti-derivative f of $F(x) = 2 \cos(2x)$.

$$f(x) = \sin 2x + k$$

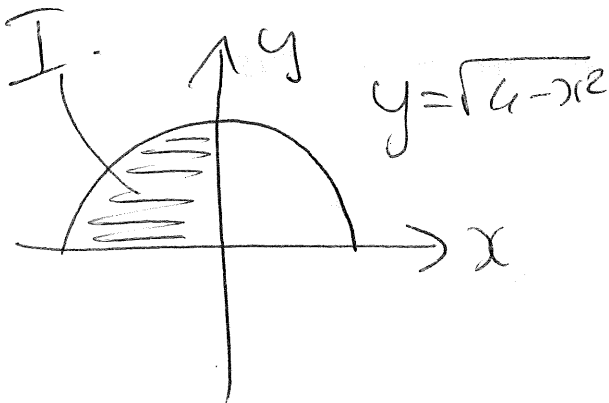
Test:

$$\frac{d}{dx}(\sin 2x + k)$$

$$= \cos 2x \times 2$$

QUESTION 2 [2 marks]

Make use of a graph and areas to find the following integral: $I = \int_{-2}^0 \sqrt{4-x^2} dx$. Clearly indicate the region represented by I on the graph.



$$I = \frac{1}{4}(\pi r^2)$$

$$= \frac{1}{4}(\pi (2)^2)$$

$$= \pi$$

QUESTION 2 [5]

Consider $f(x) = \ln(x^2 - 9)$, with $f'(x) = \frac{2x}{x^2 - 9}$, $f''(x) = \frac{-18 - 2x^2}{(x^2 - 9)^2}$. The domain of f is $(-\infty, -3) \cup (3, \infty)$ and the graph of f has no horizontal asymptotes.

Sketch the graph of f by considering all intercepts, symmetry, vertical asymptotes, intervals of increase and decrease, local extreme values and concavity.

~~See memo of test: Monday group.~~

(1) Intercepts: ~~$x=0$~~ $\Rightarrow$ No y -intercept (not defined if $x=0$)

$$y=0 \Rightarrow \ln(x^2 - 9) = 0 \\ \Rightarrow e^{\ln(x^2 - 9)} = e^0 = 1 \Rightarrow x^2 - 9 = 1 \Rightarrow x^2 = 10 \\ \Rightarrow x = \pm\sqrt{10}$$

(2) Symmetry: $f(-x) = \ln((-x)^2 - 9) = \ln(x^2 - 9) = f(x) \\ \Rightarrow f$ is an even function.

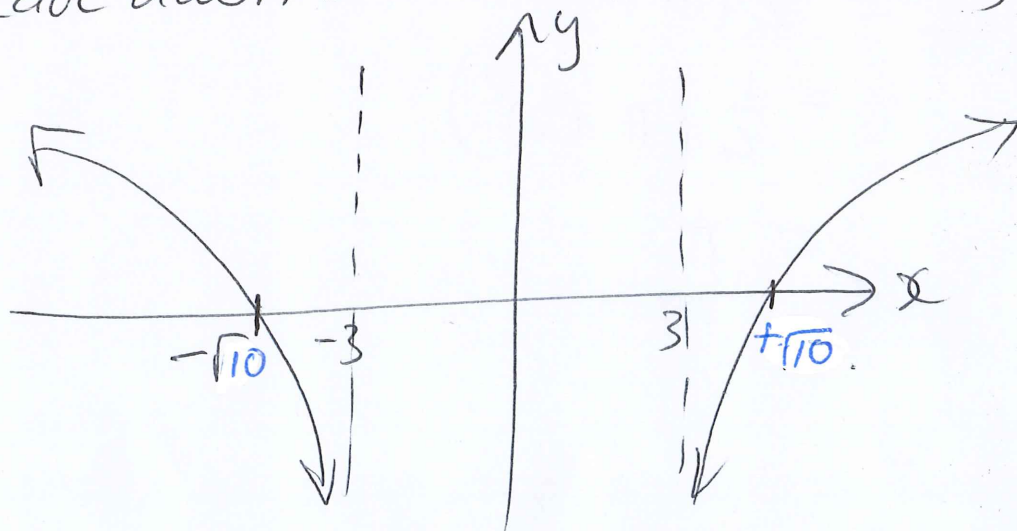
(3) Vertical asymptotes: f not defined at $x = -3$ and at $x = 3$:

$$\left. \begin{array}{l} \lim_{x \rightarrow -3^-} \ln(x^2 - 9) = -\infty \\ \lim_{x \rightarrow 3^+} \ln(x^2 - 9) = \infty \end{array} \right\} \begin{array}{l} x = -3 \text{ and } x = 3 \\ \text{are vertical asymptotes.} \end{array}$$

(4) $f'(x) = \frac{2x}{x^2 - 9} > 0$ if $x > 3 \Rightarrow$ increases on $(3, \infty)$
 $f'(x) = \frac{2x}{x^2 - 9} < 0$ if $x < -3 \Rightarrow$ decreases on $(-\infty, -3)$

(5) $f''(x) < 0 \forall x \in (-\infty, -3) \cup (3, \infty)$

$\Rightarrow f$ is concave down on $(-\infty, -3)$ and on $(3, \infty)$



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QUESTION 1 [1,1,1]

Find the most common anti-derivative f of the function given.

(1.1) $F(x) = 2 \cos(2x)$

Test: $\frac{d}{dx}(\sin 2x + c)$
 $= 2 \cos 2x$

$f(x) = \sin 2x + c$

(1.2) $F(x) = \frac{1}{2x} = \frac{1}{2} \left(\frac{1}{x} \right)$

Test: $\frac{d}{dx} \left(\frac{1}{2} \ln x + c \right)$
 $= \frac{1}{2} \times \frac{1}{x} = \frac{1}{2x}$

$f(x) = \frac{1}{2} \ln x + c$

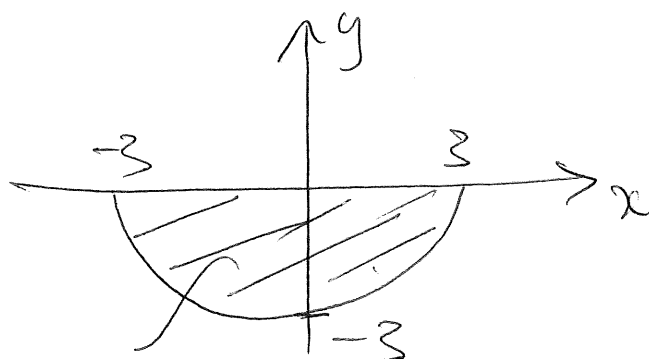
(1.3) $F(x) = \frac{x}{1+x^2} = \left(\frac{2x}{1+x^2} \right) \times \frac{1}{2}$

Test: $\frac{d}{dx} \left(\frac{1}{2} \ln(1+x^2) \right)$
 $= \frac{1}{2} \times \frac{1}{1+x^2} \times 2x = \frac{x}{1+x^2}$

$f(x) = \frac{1}{2} \ln(1+x^2) + c$

QUESTION 2 [2 marks]

Make use of a graph and areas to find the following integral: $I = -\int_{-3}^3 \sqrt{9-x^2} dx$. Clearly indicate the region represented by I on the graph.

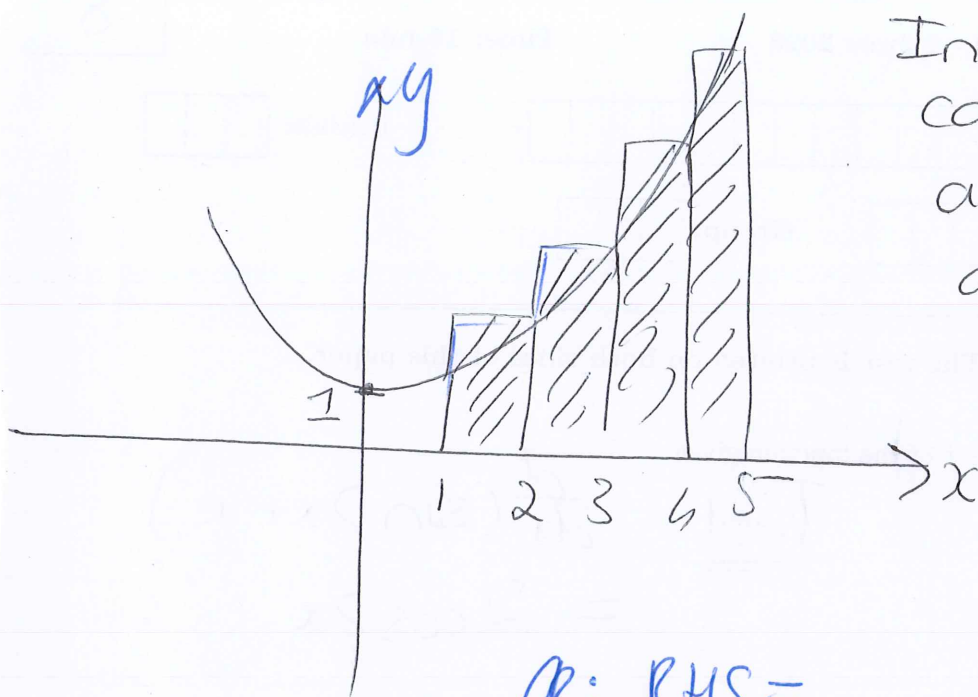


Area = $\frac{1}{2} (\pi r^2)$
 $= \frac{1}{2} \pi 9 = \frac{9}{2} \pi$

$\Rightarrow I = -\frac{9}{2} \pi$

QUESTION 3 [3 marks]

Use a right hand sum with 4 sub-intervals to estimate the area between the graph of $y = x^2 + 1$, the x -axis and $x = 1$ and $x = 5$. Show all your steps and indicate the estimated area on a graph.



Indicate if the calculated area is an over-estimate or an under-estimate.

$$\Delta x = \frac{b-a}{n}$$

$$= \frac{5-1}{4}$$

$$= 1.$$

Q: RHS = ...

Estimate of area:

$$= 1 \cdot (f(2) + f(3) + f(4) + f(5))$$

$$= 1 (4+1 + 9+1 + 16+1 + 25+1)$$

$$= 1 (5+10+17+26)$$

$$= 58$$

$$\begin{array}{r} 15 \\ 17 \\ \hline 32 \end{array}$$

$$\begin{array}{r} 32 \\ +26 \\ \hline 58 \end{array}$$

(This is an over-estimate of the area.)